

Zachary Benson

Senior full stack engineer (Ruby on Rails, React, Angular, PHP, AWS, TypeScript)

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PROFESSIONAL SUMMARY

As a highly skilled software engineer with over 8 years of experience, I specialize in building scalable, high-performance web and mobile applications, with a strong focus on full-stack development. I have extensive expertise in modern frontend technologies like **React, Angular, Vue, Next.js and TypeScript**, as well as backend development using tools such as **Ruby and Node.js**, enabling me to design and deploy robust, maintainable applications across various industries.

I am a valuable contributor to agile development teams, actively participating in architectural decisions and promoting a collaborative culture of learning and mentorship. My ability to understand complex business problems allows me to translate those needs into effective, user-centric, and technically sound solutions—working closely with product managers, engineers, designers, and data scientists across the full development stack.

With a positive attitude and a commitment to continuous learning, I stay on top of the latest trends and challenges in full-stack development. I prioritize clear communication and a customer-focused approach, making me a go-to team member in fast-paced, cross-functional environments.

EDUCATION

University of Florida - Bachelor's Degree in computer Engineering, 2014 - June 2018

TECHNICAL SKILLS

Frontend Frameworks: Specialist in building high-performance, responsive Single Page Applications (SPAs) and SSR apps using React.js, Next.js, and Angular. Proficient in optimizing Server-Side Rendering (SSR) and Static Site Generation (SSG) with accessibility and responsiveness in mind.

State Management & Data Orchestration: Proficient in designing scalable state architectures using Redux Toolkit, Context API, and RxJS for complex UIs. Experienced in implementing MVI/MVVM design patterns and intelligent "stale-while-revalidate" caching strategies to ensure real-time data consistency.

Backend & Microservices Development: Advanced expertise in Ruby on Rails and Node.js/TypeScript (NestJS, Express.js) for building resilient, enterprise-grade backends. Expert in architecting scalable systems using Microservices, BFF (Backend-for-Frontend), and Domain-Driven Design (DDD) patterns.

API Architecture & Communication: Architected and consumed RESTful, GraphQL, and gRPC APIs to facilitate robust communication between distributed services. Integrated WebSockets for real-time streaming and OpenAPI/Swagger for comprehensive documentation and testing.

Cloud & DevOps Orchestration: Deep hands-on experience managing fault-tolerant ecosystems on AWS (EKS, Lambda, S3, RDS, DynamoDB) and GCP (GKE, Apigee, BigQuery). Proficient in Terraform for Infrastructure as Code (IaC) and automating delivery through ArgoCD, GitHub Actions, and Jenkins.

Messaging & Event Streaming: Expert in utilizing Apache Kafka, RabbitMQ, and Amazon SNS/SQS to enable asynchronous communication, decoupling, and real-time data streaming in distributed microservices systems.

Database Systems & Persistence Strategy: Specialist in designing and optimizing relational schemas (PostgreSQL, MySQL) and NoSQL solutions (MongoDB, DynamoDB). Expert in implementing multi-tier caching with Redis to reduce database contention and optimize system throughput.

Mobile Engineering & Hybrid Ecosystems: Specialist in cross-platform development using React Native and Flutter. Expert in architecting scalable mobile applications while leveraging Native Modules to bridge the gap between hybrid layers and underlying Android/iOS hardware APIs.

Testing & Quality Assurance: Committed to TDD/BDD excellence using RSpec (Ruby), Jest, Cypress, and Mocha. Experienced in achieving 100% test coverage for mission-critical services and setting up automated contract testing with Pact.

Security & Identity Management: Implemented enterprise-grade security protocols including OAuth 2.0, JWT, and OpenID Connect. Expert in enforcing RBAC, CORS, rate-limiting, and securing cloud environments using AWS Cognito.

Performance Optimization: Skilled in lazy loading, code splitting, and tree shaking on the frontend, alongside server-side caching (Redis, Varnish) and web vitals monitoring to ensure top-tier performance and responsiveness.

Agile Practices & Leadership: Strong background in leading Agile/Scrum environments, conducting high-level architecture reviews, and mentoring junior engineers in SOLID design patterns and Clean Code. Collaborated via JIRA, Confluence, Slack, and Figma across globally distributed teams.

WORK EXPERIENCE

Deloitte (Remote)- Ruby on Rails Full Stack Engineer

Jul 2024 – Present

- **Architected and delivered a high-performance, cloud-native e-commerce platform** using **Ruby on Rails** and **PostgreSQL**, utilizing a **service-oriented architecture** to ensure scalability, modularity, and independent deployments.
- **Led the design and architecture of RESTful and GraphQL APIs** using **Rails** and the **GraphQL-Ruby** gem, establishing consistent patterns for versioning, error handling, and authentication via **Devise**, **JWT**, and **OAuth2**.
- **Created a centralized API Gateway pattern** with rate limiting and request validation using **Rack::Attack** and **Redis**, significantly improving latency and throughput across distributed backend services.
- **Designed and built a highly scalable React component architecture** using TypeScript, following atomic design principles and Storybook-driven development, seamlessly integrated with the Rails asset pipeline via **Webpacker/Shakapacker**.
- **Implemented Next.js SSR/SSG strategies** alongside Rails-based **Turbo/Hotwire** modules to enhance frontend performance, Core Web Vitals, and SEO readiness for global e-commerce traffic.

- **Defined and maintained backend modules with Rails**, applying **Domain-Driven Design (DDD)**, specialized **Service Objects**, and **Active Record**, improving maintainability and strict separation of concerns.
- **Integrated secure payment workflows with Stripe and PayPal**, ensuring PCI-DSS compliance, and built extensible **Rails Middleware** and **Active Job** consumers for webhook handling and fraud detection.
- **Built and deployed cloud infrastructure using AWS (Elastic Beanstalk, S3, RDS, Lambda) and Terraform** for Infrastructure as Code (IaC), supporting high availability and disaster recovery.
- **Automated full CI/CD pipelines using GitHub Actions**, utilizing **Docker**, **Buildpacks**, and **Amazon EKS (Kubernetes)** to enable canary releases and environment parity.
- **Implemented full-stack observability using New Relic, Prometheus, Grafana, and the ELK Stack**, establishing standardized dashboards and alerts for deep production insights.
- **Applied Test-Driven Development (TDD) using RSpec, FactoryBot, Capybara, and Cypress**, ensuring robust test coverage across unit, integration, and E2E flows.
- **Collaborated with designers to translate Figma wireframes** into responsive, accessible, and scalable UI systems, promoting consistent branding across platforms.
- **Authored technical documentation, API specifications (Swagger/rswag), and DevOps runbooks** to support ongoing development and team knowledge sharing.
- **Mentored frontend and backend engineers on Clean Architecture**, code quality, and **Agile/Scrum delivery practices**, contributing to a high-performing, cross-functional team.

General Motors (Florida) – Java Full Stack Engineer

Oct 2022 - May 2024

- **Architected and developed a real-time vehicle tracking platform** using **Ruby on Rails** and **PostgreSQL**, supporting thousands of concurrent users and delivering accurate live telemetry and fleet performance insights.
- **Designed and deployed cloud-native infrastructure on AWS**, leveraging **S3, Lambda, API Gateway, and Amazon Cognito**, ensuring secure authentication and highly available backend services.

- **Engineered real-time data streaming systems using Action Cable (WebSockets)**, enabling instant map updates, route tracking, and telemetry broadcasting with low-latency data handling in a distributed environment.
- **Developed secure authentication and authorization layers** using **Devise**, **OAuth2**, **JWT**, and Multi-Factor Authentication (MFA), enhancing system resilience and enterprise-grade data protection.
- **Built scalable, reusable UI components and Directives with Angular + TypeScript**, implementing advanced state management patterns with **NgRx** and **Services** to ensure consistent and performant frontend behavior.
- **Designed and implemented RESTful APIs and GraphQL endpoints (GraphQL-Ruby)** for modular data consumption, aligning backend services with scalable **Service-Oriented Architecture (SOA)** models.
- **Optimized PostgreSQL performance** using advanced indexing, partitioning, execution plan analysis, and **Active Record**, improving data retrieval speeds and reducing query load under high traffic.
- **Implemented CI/CD pipelines using Jenkins, GitLab CI, and Docker**, ensuring automated testing, build, and zero-downtime deployment via **Kubernetes (EKS)** across staging and production environments.
- **Enabled infrastructure-as-code (IaC) practices using Terraform and AWS CloudFormation**, standardizing environment provisioning and improving deployment consistency across the stack.
- **Led development of interactive dashboards** using custom charting libraries and Map APIs, utilizing **Angular's change detection strategies** and **RxJS Observables** for high-frequency visual analytics.
- **Applied performance tuning techniques** such as **Sidekiq** for asynchronous background processing, **Redis** for distributed caching, and Angular lazy loading to significantly reduce initial load times.
- **Authored and maintained comprehensive technical documentation**, API references, architectural diagrams, and operational guides for development and DevOps teams.
- **Championed Agile and Scrum methodologies**, conducting sprint planning and code reviews while mentoring junior engineers in **SOLID principles**, **Clean Architecture**, and Ruby best practices (following the Rails Way).
- **Conducted extensive automated and manual testing using RSpec, FactoryBot, Jasmine, and Karma**, ensuring robust QA coverage across backend and frontend services.

Zillow (Florida) – Java Full Stack Engineer,

Mar 2020 - Sep 2022

- Worked with a talented team to build high-performance web and mobile applications using **Ruby on Rails** for clients in retail, fintech, and healthcare.
- Focused on enhancing user experiences and optimizing **full-stack performance** by leveraging **Rails MVC architecture**, **Hotwire (Turbo/Stimulus)**, and **TypeScript**.
- Led development for various web and mobile applications, utilizing **Ruby on Rails** as a robust API backend for **React Native** clients across retail, fintech, and healthcare industries.
- Developed user-friendly, responsive UIs for large-scale projects like **Rails-based ERP systems** and custom dashboards for real-time data visualization using **ActionCable**, improving overall UX and performance.
- Collaborated with cross-functional teams to deliver highly performant, scalable solutions, optimizing **Active Record queries** and reducing server response times by 20%.
- Played a key role in optimizing build processes, reducing deployment time by 15% through **Dockerized Rails environments** and **CI/CD pipeline** enhancements.
- Integrated third-party APIs and services into the **Rails ecosystem**, ensuring seamless functionality across various platforms, including CRM and HRM systems.
- Implemented A/B testing and data-driven recommendations within the **Rails application layer** that boosted conversion rates by up to 15% for key client projects.
- Utilized Agile methodologies and tools like JIRA and Git for efficient project management, ensuring timely delivery of **Ruby gems** and features.
- Mentored junior engineers, providing guidance on best practices in **Ruby design patterns**, **RSpec testing**, and improving team performance.
- Developed and maintained reusable **Ruby libraries and UI component patterns**, promoting consistency and accelerating development across projects.
- Worked closely with UX/UI designers to ensure designs were accurately implemented while improving **server-side rendering** performance and responsiveness.
- Participated in client meetings and presentations, offering technical insights on **Rails scalability** to help drive product decisions and ensure alignment with business goals.

TePe USA (California) –Ruby on Rails Developer

Sep 2018 – Feb 2020

- Architected high-scale enterprise platforms using **React JS and TypeScript**, focusing on reusable UI components and **modular Rails-integrated design patterns**.
- Leveraged **Tailwind CSS** for consistent styling; engineered complex Menus and Drawer components using **TypeScript** for strict type safety and **ViewComponent** for Ruby-side encapsulation.
- Integrated the **html2canvas** npm package to provide advanced client-side control over image rendering and screen capture functionality within a **Rails/Turbo** environment.
- Implemented robust identity management and secure user authentication flows utilizing **Devise** and **OmniAuth** for seamless OAuth integration.
- Engineered high-performance backend services using **Ruby on Rails API mode**, adhering to **Domain-Driven Design (DDD)** and **SOLID** principles for clean separation of concerns.
- Managed relational data schemas with **PostgreSQL** and utilized **Active Record** for database control, migrations, and object-relational mapping.
- Streamlined data transformation using **ActiveModel::Serializers** or **Jbuilder**; integrated **Rswag** to generate interactive **Swagger UI** documentation for the Rails API.
- Conducted rigorous code quality audits and performance profiling using **RuboCop** and **Rack::MiniProfiler** to optimize backend and frontend delivery.
- Managed the full deployment lifecycle for frontend and Rails projects hosted on **DigitalOcean**, utilizing **App Platform** or **Droplets** for high availability.
- Applied **TDD principles** using **RSpec** and **Capybara** to ensure high code coverage for critical business logic and system integration.